HW CNS1.2 - The Passive Membrane

Neuronal Dynamics – Exercises 1.2 = Homework 1.1

General Equation

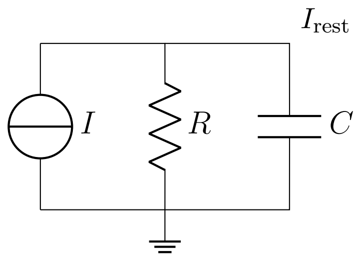

Figure 1: *
Passive membrane RC circuit

The general equation (with $V = u - u_{\text{rest}}$):

$$C\,\frac{du}{dt} = -(u - u_{\text{rest}}) + R\,I(t) \qquad \Longrightarrow \qquad \tau\,\frac{dV}{dt} = -V + R\,I(t)$$

Case i) Step Current Input

Qualitative features

Before the timestamp of step current, voltage will remain *constant* and after it will *exponentially increase* toward RI_0.

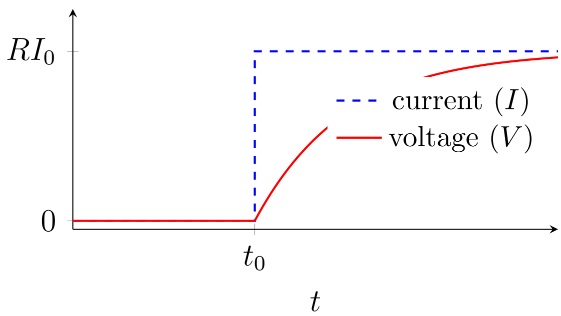

Figure 2: *
Step current (blue, dashed) and voltage response (red)

Differential Equation Derivation

i) Before step function time $(t < t_0,\ I = 0)$**:**

$$\tau \frac{dV}{dt} = -V + R\,\underbrace{I(t)}_{=0}$$

Two sub-cases:

If $V(0) = 0$**:**

$$\frac{dV}{V} = \frac{-dt}{\tau} \;\Rightarrow\; \ln V = \frac{-t}{\tau} \;\Rightarrow\; V = V_0\, e^{-t/\tau}$$

With $V(0) = 0 \;\Rightarrow\; V(t) = 0$

Equivalently:

$$\int \frac{dV}{V} = \int \frac{R\,I(i)}{\tau}\, dt \;\Rightarrow\; dV = 0$$

$\Rightarrow$ V remains constant

ii) Step current hits $(t \geq t_0,\ I = I_0)$**:**

$$\tau \frac{dV}{dt} = -V + R\,I(t)$$

Substitute $W = V - RI_0$ (so $dV = dW,\ W_0 = V(t_0) - RI_0 = -RI_0$):

$$\begin{aligned} \tau \frac{dW}{dt} &= -W & \Rightarrow \quad & \frac{dW}{-W} = \frac{dt}{\tau} \\ \Rightarrow \quad \int_{W_0}^{W(t)} \frac{dW}{W} &= \int_{t_0}^{t} \frac{dt}{\tau} & \Rightarrow \quad & -\ln\left(\frac{W(t)}{W_0}\right) = \frac{t - t_0}{\tau} \\ \Rightarrow \quad W(t) &= W_0\, e^{-(t-t_0)/\tau} & & (W_0 = -RI_0) \end{aligned}$$

Back-substitute $V(t) = W(t) + RI_0$:

$$\boxed{V(t) = RI_0\big(1 - e^{-(t-t_0)/\tau}\big)\,, \qquad t \geq t_0}$$

Case ii) Pulse Current Input

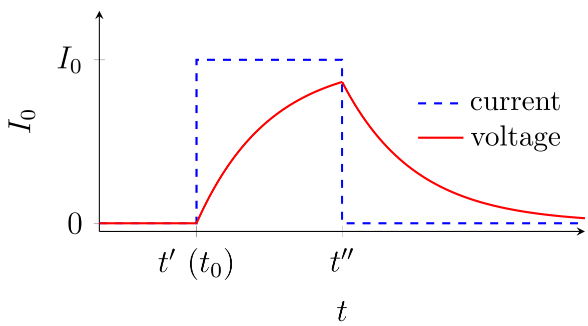

Figure 3: *
Pulse current and voltage response. Labels: $t' = t_0,\ \ t'' =$ end of pulse, $\Delta t = t'' - t'$

Qualitative feature

The whole curve can be broken down in **3 phases**: pre-pulse stage, pulse current, and post-pulse current. First two phases are analogous to the two phases of Case i. At the end of the 3rd phase, our V is approaching RI_0, but while approaching I_0, current supply is cut down, so our voltage will decrease. We have to find the equation for that.

Phase 3 – After pulse **($t > t''$, $I = 0$):**

$$\tau \frac{dV}{dt} = -V \qquad \Rightarrow \qquad \frac{dV}{V} = \frac{-dt}{\tau} \qquad \Rightarrow \qquad \ln\left(\frac{V(t)}{V''}\right) = -\frac{t - t''}{\tau}$$

where $V'' = V(t'')$ is the potential at the end of the pulse (depends on time interval of pulse):

$$V'' = RI_0\left(1 - e^{-\Delta t/\tau}\right), \qquad \Delta t = t'' - t'$$

$$\boxed{V(t) = V''\, e^{-(t-t'')/\tau} = RI_0\left(1 - e^{-\Delta t/\tau}\right)\, e^{-(t-t'')/\tau}, \qquad t > t''}$$

Case iii) Arbitrary Current Output

We *can't* split into constant pieces, so we use the **integrating factor method** on the full ODE.

Step 1 – Standard linear form

$$\frac{dV}{dt} + \frac{V}{\tau} = \frac{R}{\tau}\, I(t) \qquad \left(\text{standard: } \frac{dV}{dt} + p(t)V = q(t)\right)$$

Step 2 – Integrating factor

$$\mu(t) = e^{\int (1/\tau)\, dt} = e^{t/\tau}$$

Step 3 – Multiply both sides by μ

$$e^{t/\tau} \frac{dV}{dt} + e^{t/\tau} \frac{V}{\tau} = \frac{R\, I(t)}{\tau}\, e^{t/\tau}$$

$$\Rightarrow \qquad \frac{d}{dt}\left(e^{t/\tau}\, V\right) = \frac{R\, I(t)}{\tau}\, e^{t/\tau}$$

Step 4 – Integrate both sides

$$e^{t/\tau}\, V(t) - V_0 = \frac{R}{\tau} \int_0^t e^{s/\tau}\, I(s)\, ds$$

Final result

$$\boxed{V(t) = V_0\, e^{-t/\tau} + \underbrace{\frac{R}{\tau}\, e^{-t/\tau} \int_0^t e^{s/\tau}\, I(s)\, ds}_{\text{accumulated input, each weighted by how long ago it arrived}}}$$